UNIVERSITY OF RUHUNA

BACHELOR OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT4163 - Digital Image Processing

Practical 06 – Spatial Filtering

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1. Select a suitable grayscale image from the internet and perform the following operations.

    a) Create a variable named "Image" and read the image to Google Colab environment using open CV.
    b) Develop the averaging filter using 3X3 mask.
    c) Initializes an empty image (new_image) of the same size as the original image to store the filtered result.
    d) Apply the 3X3 mask over the image.
    e) Convert values of the result to unsigned 8-bit integers (0–255)
    f) Save the filtered image using the name "Averaging Filtered Image.tif".
    g) Display the original image and the filtered image using matplotlib.

2. Select a suitable color image from the internet and perform the following operations.
    a) Create a variable named "image" and read the image to Google Colab environment using open CV.
    b) Convert the image to RGB for correct color display.
    c) Create a kernel with the size 31.
    d) Allow sigma to automatically calculate its optimal value based on the kernel size.
    e) Apply Gaussian filter on the image.
    f) Display the original image and the filtered image using matplotlib.

3. Select a suitable image from the internet and perform the following operations.

    a) Create a variable named "ImageLF" and Read the image to Google Colab environment using open CV.
    b) Convert image to grayscale.
    c) Apply the Laplacian filter on the image.
    d) Display the original image and the filtered image using matplotlib.

4. Select a suitable image from the internet and perform the following operations.

    a) Create a variable named “ImageOS” and read the image to Google Colab environment using open CV.
    b) Apply the median filter by traversing every 3X3 area.
    c) Save the filtered image using the name “Median Filtered Image.tif”.
    d) Display the original image and the filtered image using matplotlib.